

NULL Values & 3-Valued Logic

Q. Explain the concept of NULL in SQL. How do we check for NULL values, handle them in expressions, and how do they affect comparisons and aggregate functions

[DEF] Definition to Remember

NULL in SQL is a special marker that represents missing, unknown, or inapplicable data. It is NOT the same as zero (0) or an empty string (""). NULL literally means "no value." SQL uses Three-Valued Logic (True, False, Unknown) when dealing with NULL in comparisons.

1. Key Rules About NULL

- * NULL != 0 (zero is a number; NULL is the absence of a value)

2. Checking for NULL Values

You cannot use = NULL or != NULL -- these always return UNKNOWN. Use specialized operators:

```
-- IS NULL: find rows where a column has no value
SELECT * FROM Employee WHERE Manager_ID IS NULL;

-- IS NOT NULL: find rows where a column has a value
SELECT * FROM Employee WHERE Phone IS NOT NULL;
```

3. Three-Valued Logic

SQL uses True, False, and Unknown (instead of just True/False):

Condition	Result
`Salary = NULL`	**Unknown** (not False!)
`NULL = NULL`	**Unknown** (not True!)
`IS NULL` check	Correct way -- returns True/False

4. Handling NULL with COALESCE()

COALESCE(val1, val2, ...) returns the first non-NULL value in the list -- used to substitute a default value for NULLs.

5. Effect on Aggregate Functions

- * SUM(), AVG(), MAX(), MIN() -> automatically ignore NULL rows.

Example: Salaries = [100, 200, NULL] -> AVG(Salary) = (100+200)/2 = 150 *(NULL ignored)*

[MUST-WRITE] Must-Write Points (for 10 marks)

1. NULL represents missing or unknown data -- NOT zero or empty string.
2. You cannot use = NULL or != NULL; must use IS NULL or IS NOT NULL.
3. Any arithmetic expression with NULL -> result is NULL.
4. Any comparison with NULL -> result is UNKNOWN (Three-Valued Logic).
5. COALESCE(col, default) replaces NULL with a specified default value.
6. Aggregate functions (SUM, AVG, MAX, MIN) ignore NULLs automatically.

7. COUNT(*) counts all rows; COUNT(column) excludes NULL rows.

[RECALL] Quick Recall

NULL = Unknown/Missing -> != Zero/Empty String -> IS NULL / IS NOT NULL -> Arithmetic->NULL, Comparison->UNKNOWN -> COALESCE (default) -> Aggregates ignore NULLs (except COUNT*)